Domain 01 :

Shared Responsibility Model :-

- This is one of the most tested topic in exam.

Analogy :- Renting an Apartment.

- The landrod handles the building, the walls, the main gate, the water pipes.
- You handle your door lock, who you give keys to, and what you keep inside.

If a thief walks in because you left door open that's not landlord's fault.

Security of the cloud => Aws.

- Physical Data Centers, hardware, networking, the virtualization Layers.

Security In the Cloud ⇒ You

Your Data, IAM users and permissions, passwords, MFA, Firewall rules, (Security Groups) OS patches on Ec2 encryption choices.

It Shift with the Service type :-

- Ec2 ⇒ you rent a virtual Server : you patch the OS. More Responsibility on you.
- RDS ⇒ managed Database ⇒ Aws patches the OS and Database Software. you still manage the Data and who can access It.
- S3 / Lambda ⇒ Fully managed ⇒ Aws handles even more. you still Control Permissions and Data.

Scanned with CamScanner

Identity access Management (IAM)

IAM Controls who Can do what In your Aws Account. Most Real world breaches Comes from IAM. So, this is the Core to your Carrer.

Analogy :-

A office building with their own ID card System.

- user = Emp with their own ID ——— card
- Group = A department
- Role = The Rules written on the Card
- policy = A temporary badge given to an employee or Service for a Specific Job, no permanent password.

Key Concepts :-

1- Root user ⟹ the account Owner, with unlimited power. Use it only for the

task. Lock it away with MFA. Use IAM for daily tasks.

ii- Least Privilege => Give only minimum permission needed. A developer who needs access to S3 should not have the access of billing.

iii- MFA =>

password + 2nd Proof

- Even the password is leak, the attacker is stuck.

iv- policies are json documents that say Allow or deny for specific actions on specific resources. By Default everything is denied untill you allow it.

v- Access keys = A username/password like pair for Programatic Access

vi- IAM is global not tied to one region

vii- You Can force Strong passwords and rotation for all users.